Exercise 11: Mixed Exercise

1. Factorise the following completely:

a. $x^2 - 36y^2$ $(x-6y)(x+6y)$

b. $x^8 - 1$ $(x^4+1)(x^4-1)$ $(x^4+1)(x^2+1)(x^2-1)$ $(x^4+1)(x^2+1)(x+1)(x-1)$

c. $32a^4 - 2b^4$ $2(16a^4-b^4)$ $2(4a^2+b^2)(4a^2-b^2)$ $2(4a^2+b^2)(2a+b)(2a-b)$

d. $x(2x+1) - (2x+1)$ $(2x+1)(x-1)$

e. $x(x-1) + y(1-x)$ $(x-1)(x-y)$

f. $(a+b)^2 + m(b+a)$ $(a+b)[a+b+m]$

g. $3x^2 - 11x - 20$ $(x-5)(3x+4)$

h. $6x^2 - 22x - 40$ $2(x-5)(3x+4)$

i. $x^3 + 3x^2 + 3x + 9$ $x^2(x+3)+3(x+3) = (x^2+3)(x+3)$

2. Factorise the following completely:

a. $50ax^2 - 98ay^2$ $2a(25x^2-49y^2)$ $2a(5x+7y)(5x-7y)$

b. $(x+1)^2 - 1$ $(x+1+1)(x+1-1)$ $(x+2)(x)$

c. $x^2 + 4x + 4$ $(x+2)^2$

d. $9n^2 - 25$ $(3n-5)(3n+5)$

e. $x^2(x-1) - y^2(x-1)$ $(x-1)(x+y)(x-y)$

f. $x^2 + 6x - 27$ $(x-3)(x+9)$

g. $5y^2 - 22y + 8$ $(5y-4)(5y-2)$

h. $7a^2 + 19ab - 6b^2$ $(7a-2b)(a+3b)$

i. $2x^2 + 14x + 24$ $2(x^2+7x+12)$ $2(x+4)(x+3)$

j. $n^2 + 5n - 2n - 10$ $n(n+5) - 2(n+5) = (n+5)(n-2)$

k. $4xy^2 - 16xp^2$ $4x(y^2-4p^2) = 4x(y+2p)(y-2p)$

l. $-10 + 7x - x^2$ $-(x^2-7x+10) = -(x-5)(x-2)$

m. $2(a+1)^2 + 6(a+1)$ $2[(a+1)^2+3(a+1)] = 2(a+1)(a+1+3)$ $= 2(a+1)(a+4)$

n. $9a^2 + 6ab + b^2$ $(3a+b)(3a+b)$

3. Find the length of the sides of the square, if the area is:

a.

$25a^2 + 10a + 1$

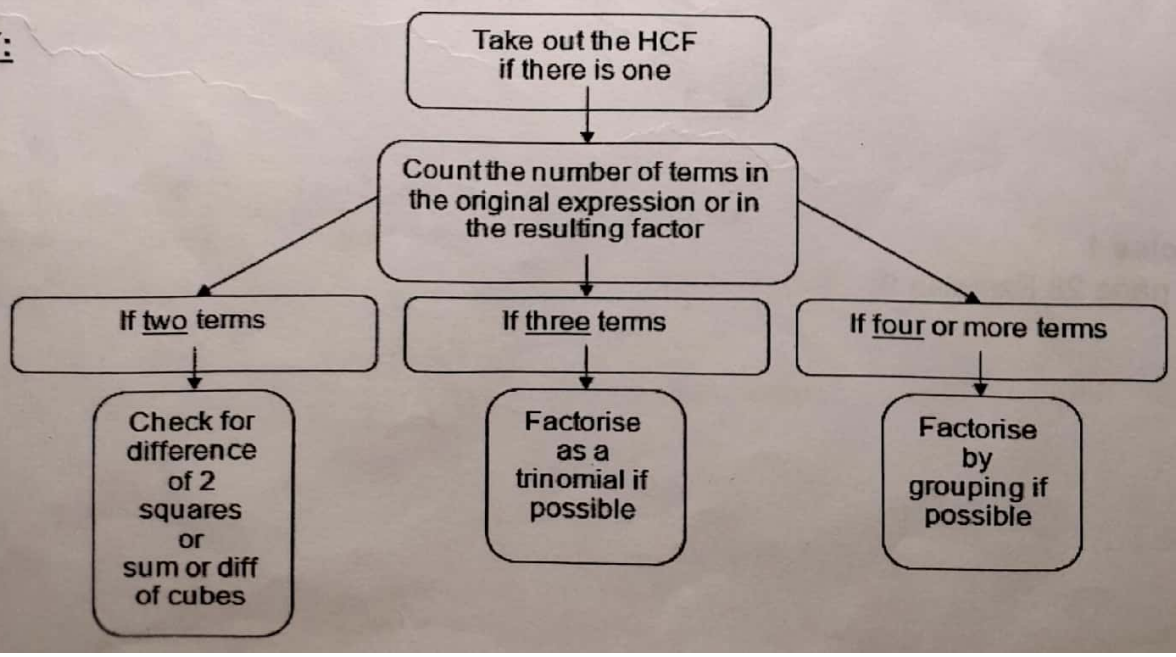
$L = (5a+1)$

b.

$y^2 + 16y + 64$

$L = (y+8)$

SUMMARY:



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